INTRODUCTION TO ROBOTICS FOR AI AND DATA SCIENCE

Topic 1: Introduction to Robotics and the AI Perspective

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YOUR MISSION: BUILD AN AUTONOMOUS SYSTEM

From Vision to Reality: A Semester-Long Journey of Incremental Integration

By the end of this course, you will have built a fully autonomous robot that can **perceive**, **localize**, **map**, **plan**, and **act**. Every lecture and lab builds one critical block of this system.

WHAT IS A ROBOT?

A programmable machine capable of carrying out complex actions automatically by **adapting** to its environment.

SENSE

PERCEPTION

Gather data about the world and internal state.

Tools: Cameras, LiDAR, IMU, Encoders

DECIDE

DECISION MAKING

Process data to plan the best action.

Tools: Algorithms, AI, Path Planning

ACT

EXECUTION

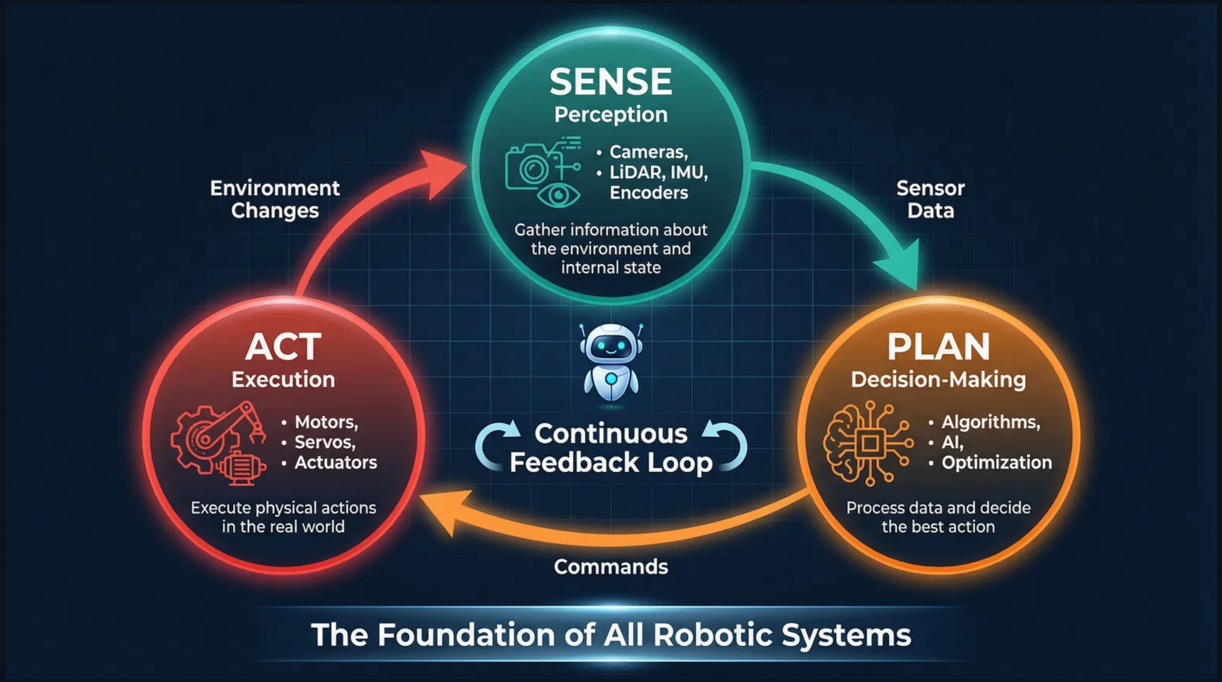
Execute physical actions to change the world.

Tools: Motors, Servos, Hydraulics

Unlike simple automation, robots operate in a Continuous Feedback Loop

THE SENSE-PLAN-ACT CYCLE

The Foundation of All Robotic Systems



SENSE

Gather data about the environment (Exteroceptive) and internal state (Proprioceptive).

PLAN

Process sensor data, update world model, and decide the optimal action.

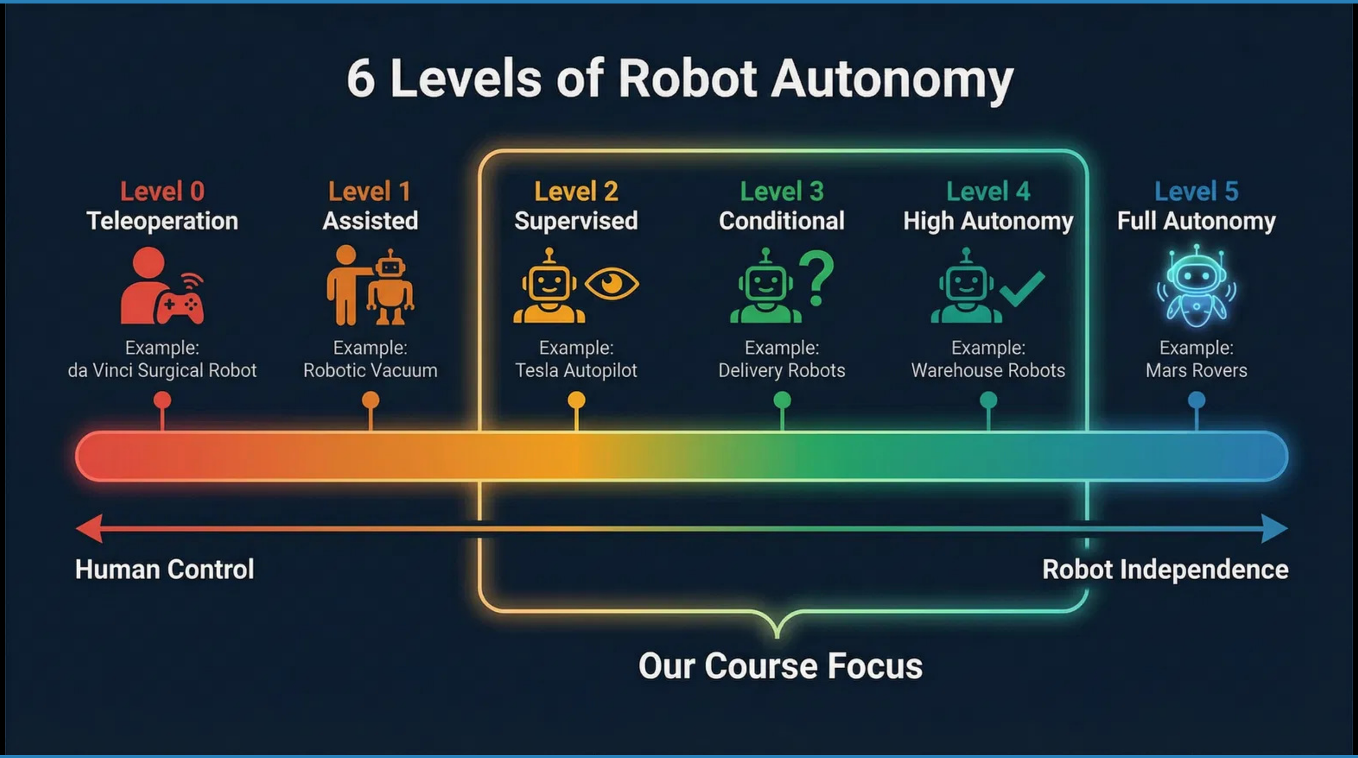
ACT

Execute commands through motors and actuators to physically change the world.

The Feedback Loop: Actions change the environment, which changes the next sensor reading.

AUTONOMY EXISTS ON A SPECTRUM

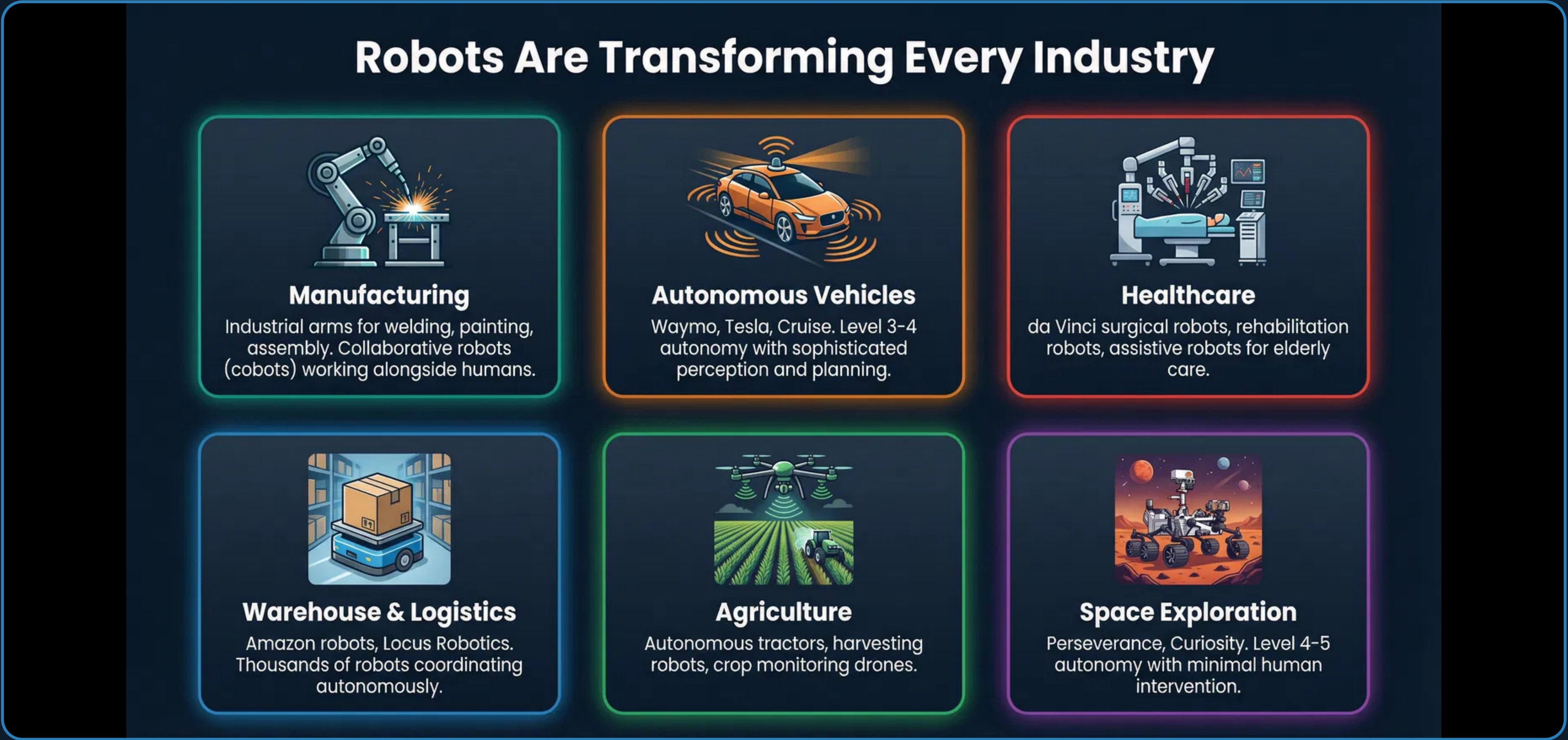
From Human Control to Full Independence



Course Focus: Levels 2-4

We build systems that operate autonomously in defined conditions (like a warehouse or a track) but may require human oversight for edge cases.

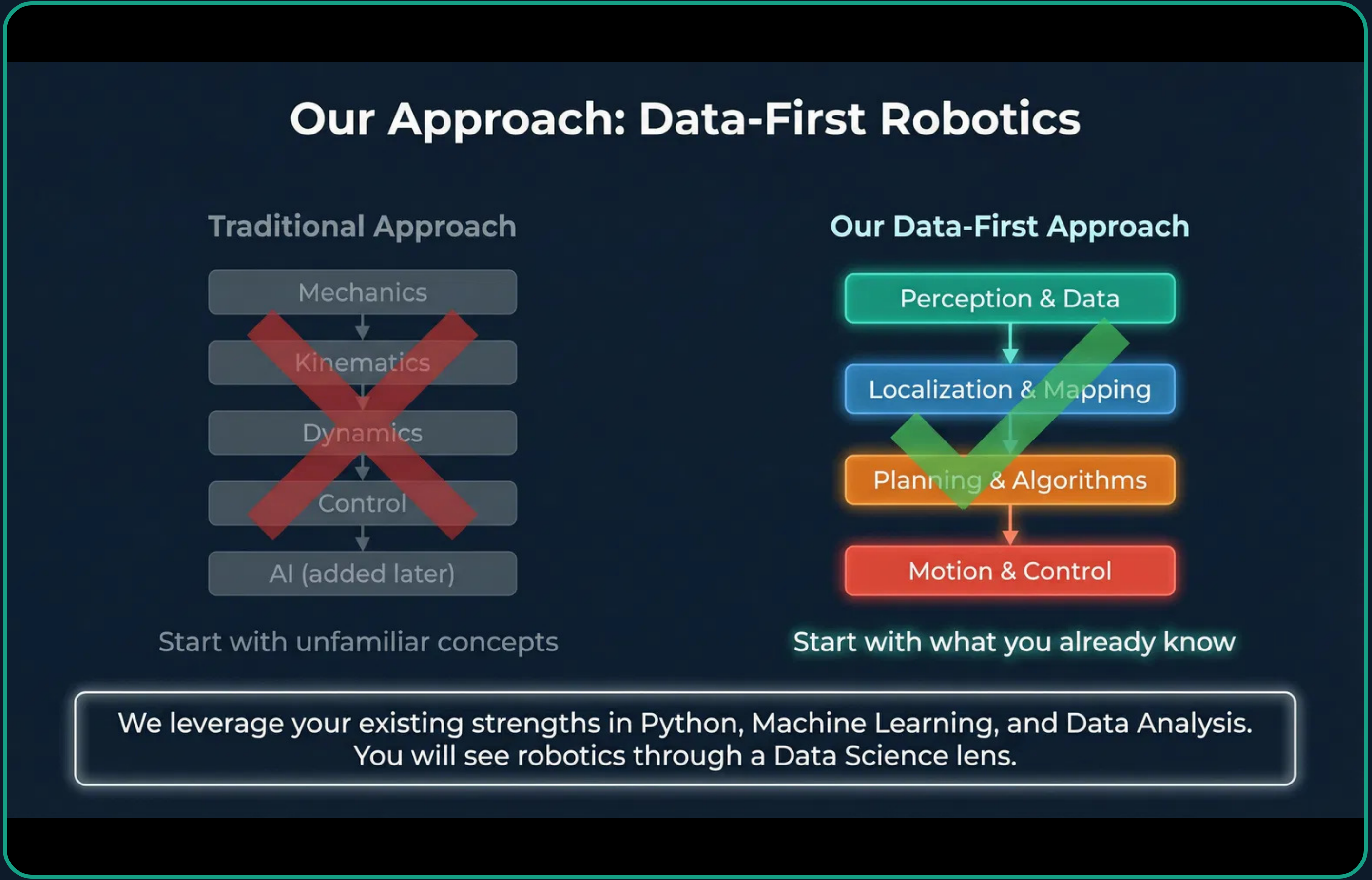
ROBOTICS IS RESHAPING INDUSTRIES



Robotics is not a niche field; it is the **physical interface of AI** transforming every major sector.

OUR DATA-FIRST APPROACH

Leveraging Your AI & Data Science Superpowers



We flip the script. Instead of struggling with mechanics first, you apply your existing **Python** and **Machine Learning** skills to perception and planning immediately.

THE INCREMENTAL INTEGRATION APPROACH

How We Build Complex Intelligent Systems

1. VISION FIRST

Start with the end goal: The Autonomous Car. Understand **WHY** before learning **HOW**.

2. DECOMPOSE

Break the complex system into manageable modules: Perception, Localization, Planning, Control.

3. BUILD BLOCKS

Master one module at a time through focused mini-projects.
(e.g., Build the Object Detector).

4. INTEGRATE

Connect the new block to your growing system. Verify interfaces and system synergy.

We avoid the "Black Box" problem. You build every part, so you understand the whole.

ASSESSMENT STRUCTURE

Your Grade Reflects What You Build

PROJECT WORK 60%

- Weekly Labs & Mini-Projects
- Midterm Project Checkpoint
- Final Integrated System

FINAL EXAM 40%

- Conceptual Understanding
- Theoretical Foundations
- Algorithm Analysis

** Note: There is no traditional Midterm Exam. It is replaced by the Project Checkpoint.*

YOUR TOOLKIT

Industry-Standard Tools for Modern Robotics

Python 3.11

The Language of AI

PyBullet

Physics Simulation Engine

NumPy

Matrix Math & Linear Algebra

OpenCV

Computer Vision & Image Processing

Matplotlib

Data Visualization & Plotting

Simulation-First: Experiment freely, fail safely, and iterate quickly.

WHAT YOU WILL MASTER

Aligned with Program Learning Outcomes

KNOWLEDGE

MK1

Simulation & Programming

Build and program robotic systems in simulation using Python and PyBullet, understanding core physics and software structure.

SKILLS

MS1

Perception Pipelines

Process sensor data from cameras and encoders to detect objects and extract key features.

MS2

Core Algorithms

Implement key robotics algorithms such as A* planning, SLAM, and PID control.

COMPETENCE

MC1

System Integration

Integrate perception, planning, and control into a functioning autonomous navigation system.

MC2

Creative Solutions

Design innovative solutions to open-ended robotics challenges.

MT1

Collaboration

Work effectively in teams to manage projects and communicate technical results.

TODAY'S LAB: MEET YOUR ROBOT

Stage 1: Setting the Foundation

The Goal:
Establish a working simulation environment and take control of your first autonomous agent.

1 **Environment Setup**
Configure Python 3.11 and install PyBullet, NumPy, and OpenCV.

2 **Spawn Robot**
Load a URDF model into the physics simulation world.

3 **Read Sensors**
Access real-time data: Position, Orientation, and Camera images.

4 **Motion Control**
Send velocity commands to drive the robot in a square pattern.

DELIVERABLES

</> Python Code · Demo Video · Brief Report

LET'S BUILD INTELLIGENT ROBOTS TOGETHER

I am here to support your learning journey

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"Whether you are struggling with a concept, excited about a project idea, or curious about research opportunities, my door is open."